Y10 Computer Science

Inside the Computer

Systems Architecture: an introduction

TEACHER COPY

Name: _____

Class: _____

Date: _____

Write your name on the front of your workbook.

Keep a pen in a different colour for the end of the lesson.

Part A: What do I already know?

Rate each part of a computer. Use pencil or your first pen colour now. Do the AFTER column at the end of the lesson in a different colour.

1 = I have never heard of it **2** = I have heard of it **3** = I can explain what it does

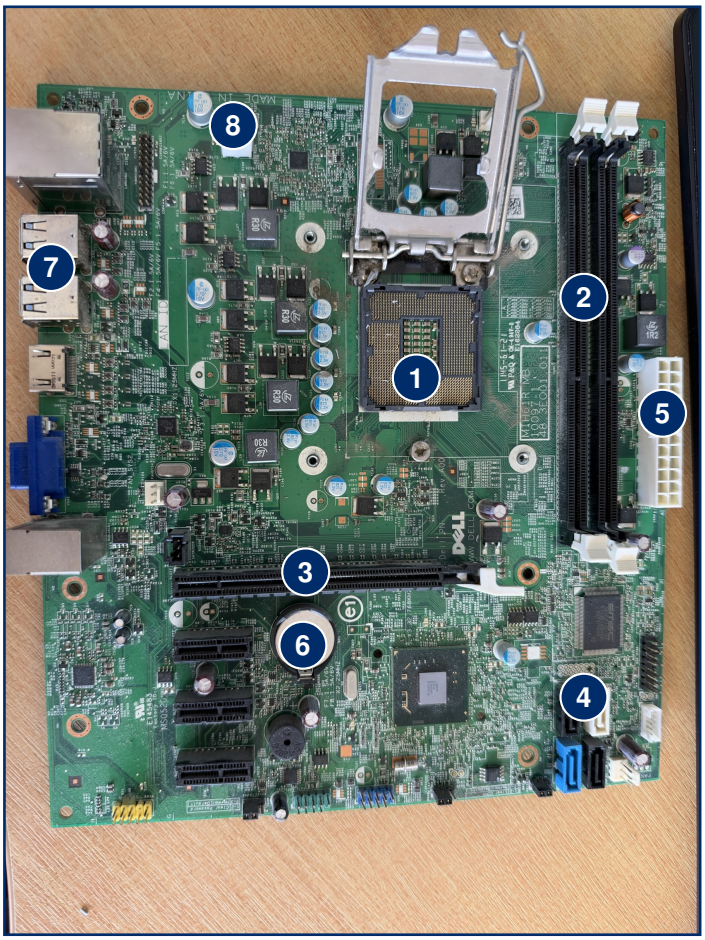
Part	**BEFORE** (1, 2 or 3)	**AFTER** (1, 2 or 3)
CPU (processor)		
RAM (memory)		
Motherboard		
Storage (hard drive or SSD)		
Power supply		
Heat sink and fan		
Graphics card		

Write one thing you already know about what is inside a computer.

Pupil's own answer. Use it to see what the class already knows.

Part B: Label the motherboard

This is the main circuit board that everything plugs into. Write the name of each numbered part.



Word bank

CPU socket · RAM slots · graphics card slot · SATA ports (storage) · main power connector · CMOS battery · back panel ports · CPU power connector

1. CPU socket
2. RAM slots
3. Graphics card slot (PCIe)
4. SATA ports (storage)
5. Main power connector
6. CMOS battery
7. Back panel ports
8. CPU power connector

Stretch: the CPU is missing from this board. Which number shows where it goes? 1

Part C: Match up

Match each part to its job, then answer the questions.

1. Match each part to its job. Write the letter in the box.

CPU	**C**
RAM	**E**
Storage (HDD or SSD)	**A**
Motherboard	**D**
Power supply	**F**
Heat sink and fan	**B**

A Keeps files and programs when the power is off

B Keeps the CPU from overheating

C Carries out instructions, like the brain

D The main circuit board that everything plugs into

E Holds the programs and data in use right now

F Changes mains electricity into power the parts can use

2. Pick up one part. Write its name and two things you notice (size, weight, metal, plastic, pins, moving parts).

Pupil's own answer, e.g. its size, weight, metal or plastic, pins, any moving parts.

3. Which part gets hot when the computer is on? How is it kept cool?

The CPU (and the graphics card). A heat sink and a fan keep it cool.

4. If you pull out the plug, one of these loses everything inside it. Which one: RAM or storage? Why do you think that is?

RAM. RAM loses everything when the power is off. Storage keeps your files without power.

Part D: Bringing it together

Fill this in as we talk.

1. The CPU repeats three steps, billions of times a second.

Word bank: RAM instruction carries out

1. **FETCH:** the CPU gets the next **instruction** from **RAM**.
2. **DECODE:** the CPU works out what the **instruction** means.
3. **EXECUTE:** the CPU **carries out** the instruction.

New word: the CPU has a **control unit**. It organises these steps, like the conductor of an orchestra. We will look inside the CPU in the next lesson.

2. How do the parts work together? Complete the sentence.

When I open a game, it is copied from **storage** into **RAM**, so that the **CPU** can run it.

3. Example: saving a document. You press Save. Fill in the gaps.

1. **FETCH:** the CPU gets the *save* instruction from **RAM**.
2. **DECODE:** the CPU works out that the document must be **written to storage**.
3. **EXECUTE:** the document is copied from RAM to **storage** so it is kept.

4. Why does the CPU need to repeat the cycle?

To keep running the program, one instruction after another, billions of times a second.

Part E: Compare two laptops

These are real laptops from a shop website. Use the table to answer the questions.



Laptop A: Dell 15



Laptop B: Dell 14 Plus

Spec	Laptop A	Laptop B
Processor	Intel Core 3 (100U)	Intel Core Ultra 7 (258V)
Processor speed	1.2 GHz	2.2 GHz
Number of cores	6	8
Cache (assume MB)	10	12
RAM	8 GB	32 GB
RAM speed	4400 MHz	8533 MHz
Storage (SSD)	512 GB	1 TB
Price	£484	£1,320

1. Which laptop has more RAM, and how many GB more? **Laptop B, 24 GB more**

2. A Blu-ray movie is 50 GB. How many could each laptop store? (Use 1 TB = 1000 GB.)

Laptop A: **10** Laptop B: **20**

3. Assume an average app uses 500 MB of RAM. How many could each laptop have open at once? (1 GB = 1000 MB.)

Laptop A: **16** Laptop B: **64**

4. Is GHz a million or a billion cycles per second? **a billion**

How many fetch-decode-execute cycles per second can Laptop A do? **1.2 billion**

Part E: Extension

Keep using the table on the previous page.

5. The RAM speed of Laptop A is 4400 MHz. Write this in GHz. **4.4 GHz**

6. Laptop B has more cores than Laptop A. How many more? Does that always mean it is that much faster? Explain.

2 more cores (8 and 6). Not always: the software has to be able to use the extra cores, and clock speed matters too.

7. Real laptops are never able to use all of their RAM for apps. Give one reason why your answer to question 3 would be lower in real life.

The operating system and other programs also use RAM.

8. A pupil only writes essays and browses the web. Which laptop would you recommend and why? Use at least two numbers from the table.

Either, with numbers. Laptop A is enough for essays and browsing (8 GB RAM, 512 GB) and costs £836 less. Laptop B is more than needed.

9. A video editor needs to open big files and run many apps at once. Which laptop would you recommend and why?

Laptop B: 32 GB RAM for many apps, 1 TB for big video files, faster processor and RAM speed.

Part F: Looking back

What did you learn today?

1. Go back to Part A and fill in the AFTER column in your second colour.

3 things I learned:

2 parts and how they connect:

1 question I still have: